

2022

## COMPUTER SCIENCE AND ENGINEERING

Paper : CSML-0902

(Probability and Statistics)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question nos. 1, 2** and **any four** from the rest.1. Answer **any five** questions :

2×5

- Can two events be simultaneously independent and mutually exclusive? Explain.
- If  $f(x) = kx^2$ ,  $0 < x < 3$ , is to be a density function, find the value of  $k$ . Find  $P(X < 1)$ .
- If  $X$  and  $Y$  are independent random variables with means 8 and 5 and variances 4 and 9, respectively, find the mean and variance of  $Z = 3X - 4Y$ .
- If  $X$  is a Poisson variate such that  $E(X^2) = 6$ , find  $E(X)$ .
- If  $X$  has uniform distribution in  $(-3, 3)$ , find  $P(|X| < 2)$ .
- Define type-I and type-II errors.

2. Answer **any five** questions :

4×5

- What is the probability that there will be 53 Sundays in (i) 1 leap year and (ii) a non-leap year?
- If the probability mass function of a random variable  $X$  is given by  $P(X = r) = kr^3$ ,  $r = 1, 2, 3, 4$ , find
  - the value of  $k$
  - $P(1/2 < X < 5/2 | X > 1)$ ,
  - the mean and variance of  $X$ .
- If the independent random variables  $X$  and  $Y$  have 36 and 16 variances, find the correlation coefficient between  $(X + Y)$  and  $(X - Y)$ .
- A point is chosen on a line of length 'a' at random. What is the probability that the ratio of the longer segment is less than  $1/4$ ?
- Find the mean and variance of a Poisson distribution with parameter 'm'.
- The following distribution of faces is obtained in 90 throws of a single die.

|             |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|
| Faces       | 1  | 2  | 3  | 4  | 5  | 6  |
| Frequencies | 15 | 19 | 13 | 12 | 16 | 15 |

Test whether the die is unbiased, i.e. the faces come with equal probabilities.

Please Turn Over

3. (a) A and B alternately throw a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, find the probability of his winning.
- (b) The chances of A, B, and C becoming the general manager of a certain company are in the ratio 4:2:3. The probabilities that the bonus scheme will be introduced in the company if A, B, and C become general manager are 0.3, 0.7 and 0.8 respectively. If the bonus scheme has been introduced, what is the probability that A has been appointed as general manager? 5+5

4. (a) A random variable  $X$  has the following probability distribution.

|        |   |     |      |      |      |       |        |          |
|--------|---|-----|------|------|------|-------|--------|----------|
| $X$    | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7        |
| $P(x)$ | 0 | $K$ | $2K$ | $2K$ | $3K$ | $K^2$ | $2K^2$ | $7K^2+K$ |

Find (i) the value of  $K$ , (ii)  $P(1.5 < X < 4.5 \mid X > 2)$ , and (iii) the smallest value of ' $m$ ' for which  $P(X \leq m) > \frac{1}{2}$ .

- (b) Buses arrive at a specified stop at 15 minutes intervals starting at 7 AM; that is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a random time that is uniformly distributed between 7 and 7:30AM, find the probability that he waits (i) less than 5 minutes for a bus, and (ii) at least 12 minutes for a bus. 5+5

5. (a) Fit a binomial distribution for the following data and hence obtain the theoretical frequencies.

|     |   |    |    |    |   |
|-----|---|----|----|----|---|
| $X$ | 0 | 1  | 2  | 3  | 4 |
| $f$ | 5 | 29 | 36 | 25 | 5 |

- (b) In a normal distribution, 7% of the items are under 35, and 89% are under 63. What are the mean and standard deviation of the distribution? What percentage of items are under 49? 5+5
6. (a) If  $T_1$ ,  $T_2$ , and  $T_3$  are independent unbiased estimators of a population parameter  $P$ , their respective variances are in the ratio of 1 : 2 : 1; which of the following estimators of  $P$  would you prefer?  $(T_1 + T_2)/2$ ,  $(T_1 + T_2 + T_3)/3$ ,  $(2T_1 + T_2 + T_3)/4$ .
- (b) The average mark scored by 32 boys is 72 with a SD of 8, while that of 36 girls is 70 with a SD of 6. Test whether the boys perform better than girls at a 5% significance level. 5+5
7. (a) In a large city A, 20% of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?
- (b) A total number of 3759 individuals were interviewed in a public opinion survey on a political proposal. Of them, 1872 were men and the rest women. A total of 2257 individuals were in favour of the proposal, and 917 were opposed to it. A total of 243 men were undecided, and 442 women were opposed to the proposal. Do you justify or contradict the hypothesis that there is no association between sex and attitude? 5+5